Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)

PointGoal2

CarGoal2

PointPush2

CarPush2

PointButton2

CarButton2

return

hard-constraint violation rate (% episodes, 95% Wilson CI)

SafeOT-typed (ours)
FDPI (constraint-based)
SAC-Lagrangian (penalty-based)
SAC + fixed penalty (penalty-based)
SDAC (multi-constraint SOTA)
SRCPO (multi-constraint SOTA)
typed Pareto frontier (soft ≤ 25)
SAC-Pen coefficient sweep
FDPI p_f sweep
SAC-Lag cost-limit sweep